Building-Level Damage Assessment

South Lebanon (issue no. 2) I GeoAI-based Rapid Assessment . July 2026

EXECUTIVE SUMMARY

This building-level damage assessment utilizes a Geospatial Artificial Intelligence (GeoAI)-enabled remote sensing workflow to rapidly detect visible conflict-related damage to buildings across South Lebanon, primarily within the geographic area south of the Litani River, with limited cadastral coverage extending north of the river (as shown in the map). This fact sheet represents the third publication in UNDP's building damage assessment series, following the issuance of the Beirut and Mount Lebanon assessment and the initial South Lebanon assessment. It constitutes an updated assessment of South Lebanon, incorporating newly acquired very high-resolution satellite imagery collected on 16 June 2026, thereby providing a more recent analysis of visible damage compared with the first South Lebanon assessment, which was based on imagery acquired on 29 April 2026. The updated imagery enables improved detection and verification of conflict-related damage and reflects changes observed on the ground since the previous publication. The assessment was conducted by the United Nations Development Programme (UNDP) and complemented by office-based visual validation performed by UNDP teams. This step involved reviewing GeoAI outputs, minimizing evident false positives, and refining the classification of visible building-level damage where feasible. Furthermore, the assessment methodology, damage classification framework, and rubble volume estimation approach are consistent with those applied in the Beirut and Mount Lebanon assessment within this series, ensuring scientific rigor, methodological consistency, and alignment with national damage assessment standards. The methodology integrates automated deep-learning segmentation models, building footprint datasets, administrative boundary layers, and expert visual interpretation. It relies on very high-resolution optical satellite imagery, comparing a baseline layer dated 23 October 2025 with conflict imagery acquired on 16 June 2026 at the building-footprint level. This approach enables the identification of visible surface-level damage indicators, including roof collapse, structural deformation, and debris accumulation. Debris volume and weight were estimated by combining building footprint areas with floor-count data derived from the Global Human Settlement dataset of the European Commission. Minor damage was excluded from the debris calculations. Debris weight was estimated using low- and high-density material scenarios of 1.69 t/m^3 and 2.25 t/m^3, based on coefficients adopted by UNDP and ESCWA (2025) and Tamraz, Srour, and Chehab (2012). Preliminary damage costs were calculated by applying graduated damage coefficients to a standardized replacement cost baseline of USD 450 per square meter, accounting for prolonged market inflation. Future publications will incorporate more recent satellite imagery and may expand to include additional areas of interest relevant to this report. The figures presented herein are preliminary and intended for programming purposes, pending validation through field-based assessments.

ASSESSMENT WORKFLOW

23 October 2025 to 16/20 June 2026

Satellite imagery comparison → GeoAI / deep-learning segmentation → Building footprint analysis → Office-based visual interpretation → Damage, debris and cost estimation

DAMAGE CLASSIFICATION

DESTROYED BUILDINGS (Dark red / 100%)
Complete structural collapse.

PARTIALLY DAMAGED BUILDINGS (Red / 50%)
Compromised load-bearing stability and partial collapse.

MINOR DAMAGE BUILDINGS (Yellow / 10%)
Limited and non-significant damage.

CAVEATS

Out of 183,018 building footprints in the study area, 12,129 buildings were detected as damaged in the previous conflict of 2024, and 3,596 cloud-obscured buildings were excluded (167 buildings fell in both categories). The remaining 167,460 buildings were assessed.

Coverage of the 16 June 2026 satellite imagery included a full coverage of 143 out of 163 cadastres, and 92.4% of all assessed buildings have complete footprint coverage. Twenty cadastres have incomplete coverage, most notably Khayyeh Saida (2% mean coverage), Aabba (7%), Jibchit (12%), Matariyet Ech-Choumar (12%), Marjaayoun town (13%), Nabatiyeh El-Faouka (17%), Kfar Tibnit (31%), Zrariyeh (47%), Khiyam (53%), and Aamra (53%).

Validation with the United Nations Department of Safety and Security (UNDSS) and the Lebanese Armed Forces (LAF) was waived due to the extensive level of destruction in the assessed area, as well as the high confidence in the methodology and validation process conducted by UNDP.

This assessment is not cumulative of damages incurred during the 2024 conflict; rather, it reflects conditions starting October 23, 2025 until June 16, 2026 only.

The assessment does not include damage to basements or underground areas, which means the overall level of damage may be underestimated.

Damage to important infrastructure—such as roads, bridges, electricity, water systems, and telecommunications—is not covered in this assessment.

The analysis does not clearly distinguish between different types of buildings (such as residential, commercial, or industries), making it harder to understand how different sectors are affected.

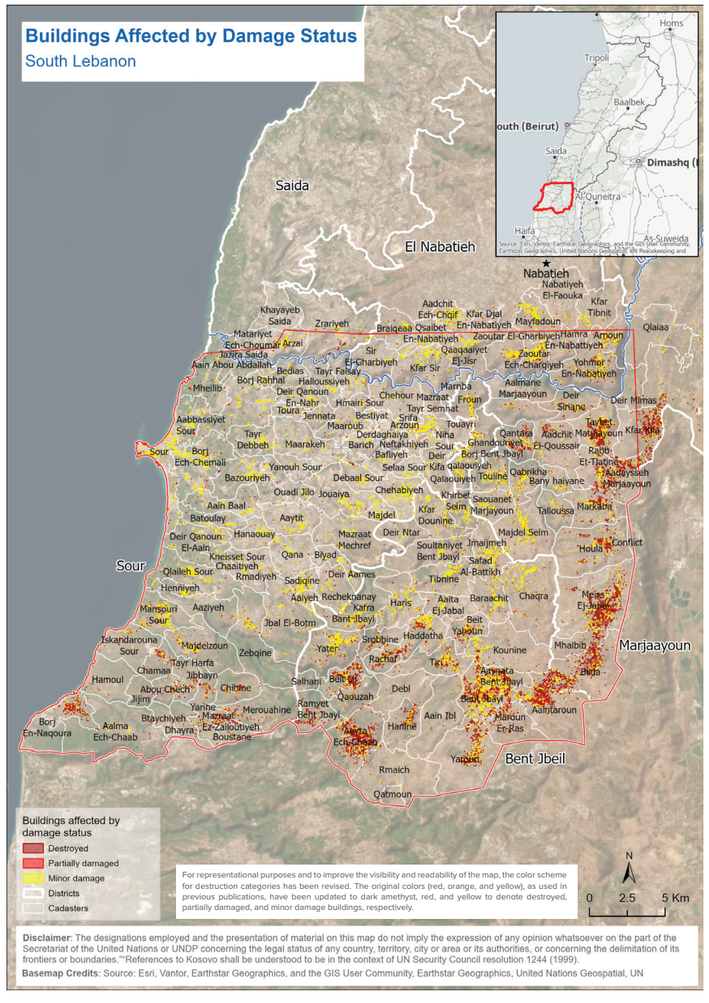
In crowded urban areas, or where buildings are close together, shadows or narrow streets can make it harder to accurately detect and classify damage.

Because the images were taken at specific points in time, some damage, repairs, or changes that happened in between may not be captured.

The assessment reflects conditions at a single moment in time and does not show ongoing damage, debris removal, or reconstruction efforts.

The South Litani assessment was validated through office-based visual interpretation only. It was not complemented by field verification or independent ground-truthing at this stage, and results should therefore serve for general understanding of the scale of damages.

DAMAGE SPATIAL DISTRIBUTION



KEY STATISTICS

24,068	7,191	21,025	7,007,132m^3	USD 3,050,134,241
Destroyed buildings	Partially damaged buildings	Minor damage buildings	Total debris volume	Total damage cost

BUILDING/UNITS DAMAGES BY GOVERNORATE

Considering an average of ~150 sqm for a single apartment

In total, 52,284 buildings (%31.2) show some degree of damage. Damage is concentrated in El Nabatieh governorate, which accounts for %85 of all destroyed buildings.

GOVERNORATE	DESTROYED	PARTIALLY DAMAGED	MINOR DAMAGED
El Nabatieh	20,568 Bldgs → 30,122 units	4,790 Bldgs → 5,903 units	12,407 Bldgs → 15,461 units
South	3,500 Bldgs → 6,017 units	2,401 Bldgs → 5,608 units	8,618 Bldgs → 17,488 units
TOTAL	24,068 Bldgs / 36,139 units	7,191 Bldgs / 11,511 units	21,025 Bldgs / 32,949 units

HOTSPOT CADASTERS

Ten most affected cadastres (by affected buildings)

1. Bent Jbayl (Bent Jbeil) — 2,712 affected (77.6% of stock) — coverage 100%
2. Aaintaroun (Bent Jbeil) — 2,528 (91.4%) — coverage 100%
3. Meiss Ej-Jabal (Marjaayoun) — 2,507 (74.4%) — coverage 100%
4. Blida (Marjaayoun) — 1,495 (87.1%) — coverage 100%
5. Taybet Matjaayoun (Marjaayoun) — 1,426 (69.1%) — coverage 100%
6. Aaynata Bent Jbayl (Bent Jbeil) — 1,285 (78.6%) — coverage 100%
7. Aayta Ech-Chaab (Bent Jbeil) — 1,276 (89.6%) — coverage 100%
8. Kfar Kila (Marjaayoun) — 1,233 (92.7%) — coverage 100%
9. Sour (Sour) — 867 (10.6%) — coverage 100%
10. Chaqra (Bent Jbeil) — 864 (26.4%) — coverage 100%

DEBRIS ESTIMATION

Minor damage excluded from debris calculations

Governorate	Volume (m^3)	Low_density weight (t)	High_density weight (t)
El Nabatieh/ Bent Jbeil	3,110,184	5,256,211	6,997,913
El Nabatieh/ Marjaayoun	1,851,462	3,128,972	4,165,790
El Nabatieh/ El Nabatieh	459,323	776,257	1,033,478
South/ Sour	1,558,068	2,633,134	3,505,652
South/ Saida	28,095	47,481	63,214
Total (South)	**7,007,132**	**11,842,055**	**15,766,047**

DAMAGE COSTS

Minor damage excluded from cost calculations

Governorate	Costs(USD)	Total (USD)
El Nabatieh/ Bent Jbeil	1,373,667,840	2,336,622,714
El Nabatieh/ Marjaayoun	769,447,015	
El Nabatieh/ El Nabatieh	193,507,859	
South/ Sour	701,861,801	713,511,527
South/ Saida	11,649,726	
Total (South)		**3,050,134,241**